

# Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11  
priors compared

31,623  
held-out plate appearances

3.05%  
held-out HR rate

0.6020  
best AUC · KNN comps

served prior

## Held-out scores

Fitted on plate appearances before 2026-08-29, scored on the 31,623 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

| PRIOR                                                                                          | AUC    | TOP-DECILE LIFT | BRIER   | LOG LOSS | CAL. RMSE |
|------------------------------------------------------------------------------------------------|--------|-----------------|---------|----------|-----------|
| <b>Flat league</b><br>League HR rate as the prior for every hitter.                            | 0.5862 | 1.61×           | 0.02950 | 0.13544  | 0.00535   |
| <b>KNN comps</b><br>Prior is the pooled rate of the nearest comparables.                       | 0.6020 | 1.55×           | 0.02945 | 0.13442  | 0.00478   |
| <b>SVR</b><br>Support-vector regression on the same feature matrix.                            | 0.5963 | 1.56×           | 0.02949 | 0.13515  | 0.00527   |
| <b>Random forest</b><br>Forest regression; a learned similarity metric.                        | 0.5956 | 1.48×           | 0.02948 | 0.13481  | 0.00643   |
| <b>Core (KNN+SVR)</b><br>Blend of KNN and SVR, no forest.                                      | 0.6020 | 1.55×           | 0.02945 | 0.13442  | 0.00478   |
| <b>Forest blend</b><br>KNN + SVR + forest, weights fitted out-of-fold.                         | 0.6020 | 1.55×           | 0.02945 | 0.13442  | 0.00478   |
| <b>Linear regression</b><br>Ordinary least squares on the same features.                       | 0.5963 | 1.51×           | 0.02947 | 0.13486  | 0.00416   |
| <b>Logistic regression</b><br>Binomial logistic on the per-PA home run outcome.                | 0.5974 | 1.55×           | 0.02949 | 0.13494  | 0.00556   |
| <b>Neural net (torch)</b><br>Two-layer PyTorch MLP, PA-weighted, trained on the same features. | 0.5964 | 1.53×           | 0.02948 | 0.13486  | 0.00505   |
| <b>XGBoost</b><br>Gradient-boosted trees, depth 3 with subsampling and L2.                     | 0.5962 | 1.55×           | 0.02949 | 0.13495  | 0.00593   |

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 252 knn 252 svr 252 rf 252 core 252 ensemble 252  
linear 252 logistic 252 nn 252 xgb 252

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,518 PA, actual HR rate 2.44%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

| PRIOR                                                                                          | AUC    | TOP-DECILE LIFT | BRIER   | LOG LOSS | CAL. RMSE |
|------------------------------------------------------------------------------------------------|--------|-----------------|---------|----------|-----------|
| <b>Flat league</b><br>League HR rate as the prior for every hitter.                            | 0.5671 | 1.39×           | 0.02386 | 0.11499  | 0.01167   |
| <b>KNN comps</b><br>Prior is the pooled rate of the nearest comparables.                       | 0.5821 | 1.39×           | 0.02386 | 0.11479  | 0.01119   |
| <b>SVR</b><br>Support-vector regression on the same feature matrix.                            | 0.5696 | 1.28×           | 0.02403 | 0.11926  | 0.01804   |
| <b>Random forest</b><br>Forest regression; a learned similarity metric.                        | 0.5845 | 1.16×           | 0.02400 | 0.11611  | 0.01452   |
| <b>Core (KNN+SVR)</b><br>Blend of KNN and SVR, no forest.                                      | 0.5821 | 1.39×           | 0.02386 | 0.11479  | 0.01119   |
| <b>Forest blend</b><br>KNN + SVR + forest, weights fitted out-of-fold.                         | 0.5821 | 1.39×           | 0.02386 | 0.11479  | 0.01119   |
| <b>Linear regression</b><br>Ordinary least squares on the same features.                       | 0.5744 | 1.05×           | 0.02396 | 0.11712  | 0.01538   |
| <b>Logistic regression</b><br>Binomial logistic on the per-PA home run outcome.                | 0.5796 | 1.28×           | 0.02395 | 0.11633  | 0.01209   |
| <b>Neural net (torch)</b><br>Two-layer PyTorch MLP, PA-weighted, trained on the same features. | 0.5761 | 1.39×           | 0.02393 | 0.11629  | 0.01479   |
| <b>XGBoost</b><br>Gradient-boosted trees, depth 3 with subsampling and L2.                     | 0.5798 | 1.16×           | 0.02403 | 0.11727  | 0.01631   |

Slate agreement

| PRIOR                | SHARED | ITS TOP SIX                                                                                                                     |
|----------------------|--------|---------------------------------------------------------------------------------------------------------------------------------|
| Served (full season) | 6/6    | Munetaka Murakami 17.8%, Miguel Vargas 15.4%, Kyle Teel 13.3%, Andrew Benintendi 13.1%, Jo Adell 13.0%, Colson Montgomery 11.9% |
| Flat league          | 5/6    | Munetaka Murakami 16.2%, Miguel Vargas 13.8%, Kyle Teel 13.2%, Jo Adell 12.3%, Andrew Benintendi 11.7%, Patrick Bailey 11.3%    |
| KNN comps            | 6/6    | Munetaka Murakami 17.8%, Miguel Vargas 15.2%, Kyle Teel 13.8%, Jo Adell 12.8%, Andrew Benintendi 12.7%, Colson Montgomery 11.9% |
| SVR                  | 5/6    | Munetaka Murakami 18.3%, Miguel Vargas 15.2%, Jo Adell 13.3%, Kyle Teel 12.7%, Andrew Benintendi 12.2%, Patrick Bailey 11.5%    |
| Random forest        | 6/6    | Munetaka Murakami 18.0%, Miguel Vargas 15.0%, Kyle Teel 13.6%, Jo Adell 13.0%, Andrew Benintendi 12.4%, Colson Montgomery 12.0% |
| Core (KNN+SVR)       | 6/6    | Munetaka Murakami 17.8%, Miguel Vargas 15.2%, Kyle Teel 13.8%, Jo Adell 12.8%, Andrew Benintendi 12.7%, Colson Montgomery 11.9% |
| Forest blend         | 6/6    | Munetaka Murakami 17.8%, Miguel Vargas 15.2%, Kyle Teel 13.8%, Jo Adell 12.8%, Andrew Benintendi 12.7%, Colson Montgomery 11.9% |
| Linear regression    | 6/6    | Munetaka Murakami 17.6%, Miguel Vargas 15.2%, Kyle Teel 13.4%, Jo Adell 13.0%, Andrew Benintendi 12.8%, Colson Montgomery 12.2% |
| Logistic regression  | 6/6    | Munetaka Murakami 17.8%, Miguel Vargas 15.4%, Jo Adell 13.3%, Andrew Benintendi 13.0%, Colson Montgomery 12.7%, Kyle Teel 12.6% |
| Neural net (torch)   | 6/6    | Munetaka Murakami 17.9%, Miguel Vargas 15.3%, Jo Adell 13.3%, Kyle Teel 12.9%, Andrew Benintendi 12.7%, Colson Montgomery 12.1% |
| XGBoost              | 6/6    | Munetaka Murakami 18.2%, Miguel Vargas 15.2%, Kyle Teel 13.5%, Jo Adell 13.2%, Andrew Benintendi 12.4%, Colson Montgomery 12.1% |

Hitter by hitter

The top 40 bats by the served probability.

| ML Model Performance Analysis: Seasonal Data |                      |             |           |       |               |                |              |                   |                     |                |  |
|----------------------------------------------|----------------------|-------------|-----------|-------|---------------|----------------|--------------|-------------------|---------------------|----------------|--|
| HITTER                                       | SERVED (FULL SEASON) | FLAT LEAGUE | KNN COMPS | SVR   | RANDOM FOREST | CORE (KNN+SVR) | FOREST BLEND | LINEAR REGRESSION | LOGISTIC REGRESSION | NEURAL NET (T) |  |
| 1Munetaka Murakami                           | 17.8%                | 16.2%       | 17.8%     | 18.3% | 18.0%         | 17.8%          | 17.8%        | 17.6%             | 17.8%               |                |  |
| 2Miguel Vargas                               | 15.4%                | 13.8%       | 15.2%     | 15.2% | 15.0%         | 15.2%          | 15.2%        | 15.2%             | 15.4%               |                |  |
| 3Kyle Teel                                   | 13.3%                | 13.2%       | 13.8%     | 12.7% | 13.6%         | 13.8%          | 13.8%        | 13.4%             | 12.6%               |                |  |
| 4Andrew Benintendi                           | 13.1%                | 11.7%       | 12.7%     | 12.2% | 12.4%         | 12.7%          | 12.7%        | 12.8%             | 13.0%               |                |  |
| 5Jo Adell                                    | 13.0%                | 12.3%       | 12.8%     | 13.3% | 13.0%         | 12.8%          | 12.8%        | 13.0%             | 13.3%               |                |  |
| 6Colson Montgomery                           | 11.9%                | 10.7%       | 11.9%     | 11.4% | 12.0%         | 11.9%          | 11.9%        | 12.2%             | 12.7%               |                |  |
| 7Patrick Bailey                              | 11.8%                | 11.3%       | 11.9%     | 11.5% | 11.5%         | 11.9%          | 11.9%        | 11.6%             | 11.3%               |                |  |
| 8Nathaniel Lowe                              | 10.9%                | 10.6%       | 11.0%     | 10.8% | 11.3%         | 11.0%          | 11.0%        | 10.8%             | 10.8%               |                |  |
| 9Chase DeLaeter                              | 10.8%                | 10.9%       | 10.7%     | 10.6% | 10.6%         | 10.7%          | 10.7%        | 10.5%             | 10.5%               |                |  |
| 10Travis Bazzana                             | 10.6%                | 10.6%       | 10.4%     | 10.2% | 10.6%         | 10.4%          | 10.4%        | 10.4%             | 10.2%               |                |  |
| 11José Ramírez                               | 9.9%                 | 9.9%        | 9.8%      | 9.6%  | 9.9%          | 9.8%           | 9.8%         | 10.0%             | 10.3%               |                |  |
| 12Tristan Peters                             | 9.8%                 | 10.4%       | 10.0%     | 10.3% | 10.2%         | 10.0%          | 10.0%        | 10.1%             | 10.2%               |                |  |
| 13Sam Antonacci                              | 9.1%                 | 10.9%       | 9.4%      | 8.5%  | 9.5%          | 9.4%           | 9.4%         | 9.3%              | 9.1%                |                |  |
| 14Chase Meidroth                             | 7.9%                 | 9.2%        | 8.3%      | 8.6%  | 8.6%          | 8.3%           | 8.3%         | 8.4%              | 8.2%                |                |  |
| 15Petey Halpin                               | 7.7%                 | 8.9%        | 7.7%      | 8.1%  | 8.3%          | 7.7%           | 7.7%         | 8.9%              | 8.3%                |                |  |
| 16Brayan Rocchio                             | 6.7%                 | 7.2%        | 6.6%      | 6.8%  | 6.4%          | 6.6%           | 6.6%         | 6.5%              | 6.5%                |                |  |
| 17Brenton Doyle                              | 6.4%                 | 8.5%        | 7.1%      | 7.2%  | 6.7%          | 7.1%           | 7.1%         | 7.0%              | 6.8%                |                |  |
| 18Steven Kwan                                | 4.6%                 | 7.8%        | 5.5%      | 4.0%  | 4.0%          | 5.5%           | 5.5%         | 3.3%              | 4.8%                |                |  |

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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Slate for 2026-10-10 · model 2026.10.01-stack · built 2026-10-11 05:40:09 · live data 2026-10-11 05:40:10

Built for research and entertainment. Probabilities are estimates, not guarantees; never stake what you cannot afford to lose. Data: MLB Stats API play-by-play, lineups and live scores; ESPN injuries and bios.